

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	23January 2026
Project Name	Metro Ticket Booking System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Metro ticket booking page with passenger details, fare calculation, and QR code display	HTML, CSS, JavaScript, Jelly (UI Page)
2.	Application Logic-1	Metro ticket booking process	Client Scripts
3.	Application Logic-2	Automatic fare calculation and QR code generation	JavaScript
4.	Application Logic-3	Booking request processing and record creation	Script Include (GlideAjax), GlideRecord
5.	Database	Stores metro booking details	ServiceNow Custom Table (u_metro_ticket_booking)
6.	Cloud Database	ServiceNow cloud-hosted database	ServiceNow Platform Database
7.	File Storage	QR code image and attachments (if required)	ServiceNow Attachments / Image Storage
8.	External API-1	QR Code Generation Library	QRCode.js
9.	External API-2	GlideAjax communication between UI Page and Script Include	ServiceNow GlideAjax API
10.	Machine Learning Model	Not Applicable	N/A

11.	Infrastructure (Server / Cloud)	Application hosted on ServiceNow cloud platform	ServiceNow Cloud Platform
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Used for dynamic QR code generation and UI scripting	QRCode.js, JavaScript
2.	Security Implementations	Role-based access, authenticated users, secure database access	ACLs, ServiceNow Security, GlideRecord
3.	Scalable Architecture	Supports multiple users and metro bookings through cloud architecture	ServiceNow Multi-Tenant Cloud
4.	Availability	Application available through ServiceNow cloud with high uptime.	ServiceNow Cloud Infrastructure
5.	Performance	Fast fare calculation, QR code generation, and booking using AJAX without page reload	GlideAjax, Client Scripts, JavaScript